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ICE FISHING HOLE COVER APPARATUS

Abstract

An ice fishing hole cover apparatus comprising a flange securable to a floor surface of a floor surrounding a floor hole with a sleeve extending from the flange into the floor hole and defining an aperture therethrough. A cover configured to biasingly magnetically engage the flange using several magnets and several corresponding metal elements disposed around the cover and around the flange respectively. The ice fishing hole cover apparatus may include an insert insertable into the sleeve in mechanical cooperation with the sleeve detent to extend the aperture to the ice fishing hole.

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Background/Summary

BACKGROUND OF THE INVENTION

Field

[0001] This disclosure relates to apparatus for ice fishing, and, more specifically, to ice fishing hole cover apparatus for selective covering of an ice fishing hole within an ice fishing house.

Background

[0002] Ice fish houses have been long used for fishing through the ice of, for example, a frozen lake or river. The ice fishing house includes a floor through which one or more floor holes are provided. The fisherman may then access the ice through the floor hole(s) in order to drill ice fishing hole(s) through the ice and then fish therethrough. Fishing may be performed, for example, by angling or by spearing through the ice fishing hole, in various aspects.

[0003] It may be important to seal a region around the ice fishing hole including the floor hole when not in use to inhibit refreezing of the ice fishing hole thereby making it easier to redrill the ice fishing hole at a later time. Prevention of snow, dirt, and debris from blowing through the floor hole and onto the ice and/or into the ice fishing hole by sealing the floor hole(s) may eliminate a nuisance. Sealing the floor hole may eliminate a hazard e.g., prevents stepping into the hole when walking about the floor of the ice fish house. Improved esthetics around the floor hole are also a consideration. Sealing the gap between the floor and the ice around the floor hole may prevent wind from blowing via the gap into the ice fishing house through the floor hole. Accordingly, there is a need for improved apparatus as well as related methods for covering selectively an ice fishing hole within an ice fishing house.

BRIEF SUMMARY OF THE INVENTION

[0004] These and other needs and disadvantages may be overcome by the apparatus and related methods of use disclosed herein. Additional improvements and advantages may be recognized by those of ordinary skill in the art upon study of the present disclosure.

[0005] An ice fishing hole cover apparatus is disclosed herein. In various aspects, the ice fishing hole cover apparatus includes a base ring having a flange securable to a floor surface of a floor surrounding a floor hole with a sleeve extending from the flange into the floor hole and defining a base ring aperture therethrough. A cover is configured to biasingly magnetically engage the flange using several magnets and several corresponding metal elements disposed around the cover and around the flange respectively to selectively cover the base ring aperture. The ice fishing hole cover apparatus may include an insert insertable into the sleeve in mechanical cooperation with the sleeve detent to extend the aperture to the ice fishing hole. The cover may include a cover insert extending forth from a cover side of the cover, and the cover insert is insertable into the base ring aperture. In various aspects, the cover insert defines an outer boundary having a frustoconical shape configured to guide the cover flange into biased engagement with the recessed surface of the flange by insertion of the cover insert into the base ring aperture.

[0006] This summary is presented to provide a basic understanding of some aspects of the apparatus and methods disclosed herein as a prelude to the detailed description that follows below. Accordingly, this summary is not intended to identify key elements of the apparatus and methods disclosed herein or to delineate the scope thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1**A illustrates by perspective view an exemplary implementation of ice fishing hole cover apparatus according to the present inventions;

[0008] FIG. **1**B illustrates by plan view the exemplary ice fishing hole cover apparatus of FIG. **1**;

[0009] FIG. **2** illustrates by exploded perspective view the exemplary ice fishing hole cover apparatus of FIG. **1**A;

[0010] FIG. **3**A illustrates by cross-sectional elevation view portions of the exemplary ice fishing hole cover apparatus of FIG. **1**A;
[0011] FIG. **3**B illustrates by cross-sectional elevation view portions of the exemplary ice fishing hole cover apparatus of FIG. **1**A; and,
[0012] FIG. **4** illustrates the exemplary ice fishing hole cover apparatus of FIG. **1**A by cross-sectional elevation view through cutline **4-4** of FIG. **1**B.
[0013] The Figures are exemplary only, and the implementations illustrated therein are selected to facilitate explanation. The number, position, relationship and dimensions of the elements shown in the Figures to form the various implementations described herein, as well as dimensions and dimensional proportions to conform to specific force, weight, strength, flow and similar requirements are explained herein or are understandable to a person of ordinary skill in the art upon study of this disclosure. Where used in the various Figures, the same numerals designate the same or similar elements. Furthermore, when the terms "top," "bottom," "right," "left," "forward," "rear," "first," "second," "inside," "outside," and similar terms are used, the terms should be understood in reference to the orientation of the implementations shown in the drawings and are utilized to facilitate description thereof. Use herein of relative terms such as generally, about, approximately, essentially, may be indicative of engineering, manufacturing, or scientific tolerances such as ±0.1%, ±1%, ±2.5%, ±5%, or other such tolerances, as would be recognized by those of ordinary skill in the art upon study of this disclosure.
DETAILED DESCRIPTION OF THE INVENTION
[0014] As illustrated in FIGS. **1**A, **1**B, exemplary ice fishing hole cover apparatus **10** includes cover **30** removably received by base ring **20**. Base ring **20**, as illustrated, is received partly within floor hole **13** formed in floor **12** with base ring **20** generally engaging floor **12** around a periphery of floor hole **13**. Fasteners, such as fasteners **27***a*, **27***b*, **27***c*, engage base ring **20** and engage floor surface **17** of floor **12** through holes, such as holes **23***a*, **23***b*, **23***c*, respectively, disposed circumferentially around base ring **20** to secure base ring **20** to floor **12**, as illustrated. Base ring **20** defines base ring aperture **22** that allows communication through floor hole **13**, in this implementation. Floor **12** may comprise a portion of a structure used for ice fishing (e.g., an ice fishing house), so that floor **12** may be generally enclosed within the structure. Floor **12** may be formed out of lumber or plywood, for example. As illustrated, floor **12** overlies ice **14**, and base ring aperture **22** received by floor hole **13** aligns with an ice fishing hole **15** (also see FIG. **4**) passing through ice **14** to allow fishing therethrough by a fisherman positioned above floor surface **17** of floor **12**. Ice fishing hole **15** may be formed by chisel or ice auger through base ring aperture **22**.
[0015] Cover **30**, as illustrated in FIGS. **1**A, **1**B, is removably received by base ring **20** to enclose selectively base ring aperture **22** thereby to enclose selectively floor hole **13** and portions of ice **14** beneath floor **12** including ice fishing hole **15**. When received by base ring **20**, cover side **37** of cover **30** as well as base ring **20** are generally flush with or low profile with respect to floor surface **17** of floor **12** so as not to form an obstruction about floor **12**. Tread **88** is configured in portions of cover side **37** of cover **30** to provides traction to a user walking upon floor surface **17** of floor **12** thereby preventing slipping on cover **30** when cover **30** is received by base ring **20**. Handle **40** is received in recess **42** formed in cover **30** to allow a user to manipulate cover **30** in order to engage cover **30** with base ring **20** or remove cover **30** from engagement with base ring **20**. By being received in recess **42**, handle **40** avoids obstructing cover side **37** of cover **30** and, thus, floor surface **17** of floor **12**. Note that, while base ring **20**, base ring aperture **22** and cover **30** have a circular shape in exemplary ice fishing hole cover apparatus **10**, base ring **20**, base ring aperture **22**, and cover **30** may assume other shapes such as oval, rectangular, or square, compatible with one another, in various other implementations. Base ring **20** and cover **30** may be generally formed of suitable injection molded plastic, as would be readily recognized by those of ordinary skill in the art upon study of this disclosure. In certain implementations, cover **30** may be configured to be

insulating, and, thus, may include, for example, foam plastic, and may have sufficient thickness to provide insulation, as would be readily recognized by those of ordinary skill in the art upon study of this disclosure.

[0016] As illustrated in FIG. **2**, base ring **20** includes sleeve **26** and flange **24**. Sleeve **26** extends forth from flange **24** and is inserted into floor hole **13** in disposition generally around the periphery of floor hole **13** with flange **24** overlying floor surface **17** of floor **12**, as illustrated. At least portions of sleeve **26** may bias against the periphery of floor hole **13** in certain implementations. Thus, sleeve **26** bounds portions of base ring aperture **22** with other portions of base ring aperture **22** being bounded by flange **24**, in this implementation. Flange **24** of base ring **20** may be secured to floor surface **17** of floor **12** by fasteners, such as fasteners **27***a*, **27***b*, **27***c*, through holes, such as holes **23***a*, **23***b*, **23***c*, respectively, provided circumferentially around flange **24**. Flange **24** includes recessed portion **28** with recessed surface **29** that is formed around an entire inner periphery thereof, and cover **30** includes cover flange **34** formed around an entire outer periphery thereof, as illustrated. Recessed surface **29** of recessed portion **28** is configured to receive cover flange surface **39** of cover flange **34** around an entire outer periphery of cover **30** in biased engagement when cover **30** is engaged with base ring **20**, as illustrated in FIG. **2** and FIG. **3**A.

[0017] As illustrated in FIG. **2**, cover **30** includes cover insert **32** that extends forth from cover side **33** of cover **30** with cover side **33** being opposite of cover side **37**. Cover insert **32** includes barrel **36** with plates, such as plates **35***a*, **35***b*, **35***c*, disposed peripherally with edges of the plates in radial alignment and axial alignment around barrel **36**, as illustrated in FIGS. **2**, **3**A. The plates are triangular in shape, as illustrated in FIG. **3**A, so that edges of the plates define outer boundary **38** of cover insert **32** having a frustoconical shape broadest at cover side **33**. Vertexes of the planar triangular shaped plates, such as vertex **61** of plate **35***a* illustrated in FIG. **3**A, define an inner boundary **57** of cover flange surface **39** of cover flange **34**, which is an outer portion of cover side **33**. The plates secured to barrel **36** as per exemplary ice fishing hole cover apparatus **10** may structurally support cover **30** without the material and associated weight of a similarly shaped structure of continuous solid construction. When received in the base ring aperture **22**, the plates contact the base ring **20** which may support structurally the base ring **20** cover **30** combination in engagement with floor **12**. The frustoconical shape of cover insert **32** may guide the insertion of cover insert **32** into base ring aperture **22**, and the frustoconical shape of cover insert **32** may aid in aligning cover **30** with base ring **20** including the alignment of cover flange surface **39** of cover flange **34** with recessed surface **29** of recessed portion **28** of base ring **20**. For example, portions of frustoconical shaped outer boundary **38** of cover insert **32** may variously bias against portions of the sleeve **26** causing repositioning of the cover **30** as the cover insert **32** is being inserted into the sleeve **26** resulting in alignment of the cover flange surface **39** of the cover flange **34** of the cover **30** with the recessed surface **29** of the recessed portion **28** of the base ring **20** when cover insert **32** is fully received. This may be useful, for example, in low light conditions to ensure proper engagement between the base ring **20** and cover **30**. When fully received in base ring aperture **22**, cover insert **32** extends into base ring aperture **22** with barrel **36** generally parallel to sleeve **26**, and plates, such as plates **35***a*, **35***b*, **35***c*, proximate vertices, such as vertex **61**, biased against edge **59** of recessed portion **28** of base ring **20** to align the cover flange surface **39** of the cover flange **34** of the cover **30** in biased engagement with the recessed surface **29** of the recessed portion **28** of base ring **20**, in this implementation.

[0018] Handle **40** is illustrated in FIG. **2** as being secured to cover **30** using fasteners **47***a*, **47***b* (also see FIG. **4**). Handle **40** may be hingedly secured to cover **30** to be selectively positioned, in various other implementations. Handle **40** may telescopically receive itself to be selectively extended/retracted, in yet other implementations.

[0019] As illustrated in FIG. **3**A and FIG. **4**, magnets, such as magnets **51***a*, **51***b*, are disposed around an entirety of recessed portion **28** of base ring **20** to engage magnetically corresponding metal elements, such as metal elements **53***a*, **53***b*, disposed around an entirety of cover flange **34** of cover **30** to hold the entirety of cover flange surface **39** of cover flange **34** engaged with the entirety of recessed surface **29** of recessed portion **28** of base ring **20**. Thus, the entire periphery of cover flange **34** of cover **20** and the entire periphery of recessed portion **28** of base ring **20** are magnetically engaged with one another, not just at certain mechanical detents at certain specific locations. Such magnetic engagement of the entirety of cover flange surface **39** of cover **30** with the entirety of recessed surface **29** of base ring **20** may thus sealingly engage cover **30** with base ring **20** thereby sealing floor hole **13** and ice **14** beneath floor **12** including ice fishing hole **15** passing through ice **14**. Such peripheral sealing between cover **30** and base ring **20** may inhibit refreezing of ice fishing hole **15** and may keep snow, dirt, etc. from floor hole **13**, ice **14** beneath floor **12**, and ice fishing hole **15**. Although only magnets **51***a*, **51***b* and corresponding metal elements **53***a*, **53***b* are illustrated in FIG. **3**A and FIG. **4** for explanatory purposes, it should be understood that many magnets, such as magnets **52***a*, **52***b*, and corresponding metal elements, such as metal elements **53***a*, **53***b*, may be disposed peripherally at regular intervals around recessed portion **28** and cover flange surface **39**, in various implementations. For example, the number of magnets, such as magnets **51***a*, **51***b*, may be generally between about 4 magnets and about 12 magnets with a corresponding number of metal elements, such as metal elements **53***a*, **53***b*. In yet other implementations, the magnet and corresponding metal element may be configured as a continuous strip disposed around the circumference of base ring **20** and cover **30**, respectively.

[0020] The metal elements, such as metal elements **53***a*, **53***b*, are magnetically attracted to the magnets, such as magnets **51***a*, **51***b*, in this implementation. In other implementations, magnets, such as magnets **51***a*, **51***b*, may be disposed peripherally at regular intervals around cover flange **34** of cover **30**, and metal elements, such as metal elements **53***a*, **53***b*, may be disposed peripherally at regular intervals around recessed portion **28** of base ring **20**. Magnets of opposite polarity may be disposed peripherally around cover flange **34** of cover **30** and peripherally around recessed portion **28** of base ring **20** to engage magnetically cover **30** with base ring **20**, in certain implementations. In exemplary ice fishing hole cover apparatus **10**, magnets, such as magnets **51***a*, **51***b*, are illustrated as disposed between recessed portion **28** of flange **24** and floor **12**, and metal elements, such as metal elements **53***a*, **53***b*, are illustrated as embedded within cover flange **34** of cover **30**. However, it should be recognized that magnets and/or metal elements may be variously disposed about recessed portion **28** of flange **24** and cover flange **34** of cover **30** in various other ways, in various other implementations.

[0021] As illustrated in FIGS. **3**A, **3**B, **4**, exemplary ice fishing hole cover apparatus **10** may include insert **46** having a cylindrical shape and defining insert passage **45** therethrough. Sleeve **26** of base ring **20** includes sleeve detent **41** configured as a lip that extends peripherally around sleeve **26**, and insert **46** includes insert detent **43** configured as a lip that extends peripherally around insert **46**, as illustrated. Insert **46** may be insertably removably held within sleeve **26** by engagement between sleeve detent **41** and insert detent **43**, as illustrated in FIG. **3**B and FIG. **4**. Note that insert **46** is sized to pass into base ring aperture **22** at flange **24** with flange **24** engaged with floor **12** in order to engage insert detent **43** with sleeve detent **41**. It should also be recognized that insert **46** may have other shapes in conformance to other shapes of sleeve **26**, in various other implementations. Also note that structure(s) other than insert **46** may be configured to engage sleeve detent **41**, in various other implementations. Further note that several inserts, such as insert **46**, may be configured to engage telescopically one another and be held extended by insert detents, such as insert detents **43**, formed at the ends thereof, in various other implementations.

[0022] The ice fishing house may rest upon supports (not shown) such as dimensional lumber (e.g., 2×4, 4×4, etc.), various blocks, bricks, etc., so that there is generally a gap **65** between floor surface **19** of floor **12** and ice **14**, as illustrated in FIG. **4**. For example, water may flow onto ice **14** from cracks and heaves in ice **14**. Gap **65** between floor surface **19** of floor **12** and ice **14** may prevent such water flows onto ice **14** from flooding floor **12** and from freezing floor **12** into ice **14**, both of which are highly undesirable. However, gap **65** provides a pathway for the communication of wind through floor hole **13**. As illustrated in FIG. **4**, insert **46** extends from sleeve **26** of base ring **20** into ice fishing hole **15** around a periphery thereof in order to block wind communication through floor hole **13** via gap **65**. Ice fishing hole **15** may then be accessed through base ring aperture **22** and through insert passage **45** with base ring aperture **22** and insert passage **45** being generally in alignment. Accordingly, insert **46** effectively extends base ring aperture **22** to ice fishing hole **15**. In this implementation, lengths of sleeve **26** and cover insert **32** are configured to allow insert **46** to remain engaged with sleeve **26** whether or not cover **30** is received by base ring **20**. FIG. **4** illustrates insert **46** engaged with sleeve **26** and cover **30** received by base ring **20**. Note that a water surface of water within ice fishing hole **15** lies slightly below an ice surface of ice **14**, as illustrated in FIG. **4**.

[0023] As use and operation of the exemplary ice fishing hole cover apparatus **10** is readily recognizable to those of ordinary skill in the art upon study of this disclosure, no further discussion of the use or operation of the exemplary ice fishing hole cover apparatus **10** is provided.

[0024] The foregoing discussion along with the Figures discloses and describes various exemplary implementations. These implementations are not meant to limit the scope of coverage, but, instead, to assist in understanding the context of the language used in this specification and in the claims. The Abstract is presented to meet requirements of 37 C.F.R. § 1.72 (b) only. Accordingly, the Abstract is not intended to identify key elements of the apparatus and methods disclosed herein or to delineate the scope thereof. Upon study of this disclosure and the exemplary implementations herein, one of ordinary skill in the art may readily recognize that various changes, modifications and variations can be made thereto without departing from the spirit and scope of the inventions as defined in the following claims.

Claims

1. An ice fishing hole cover apparatus, comprising: a base ring defining a base ring aperture and comprising a flange securable to a floor surface of a floor surrounding a floor hole formed in the floor by a plurality of fasteners and with a sleeve surrounding the base ring aperture at the flange extending from the flange into the floor hole; a recessed portion comprising an innermost portion of the flange, the recessed portion passing around an entire periphery of the base ring aperture and defining a recessed surface; a cover having a cover flange passing around an entirety of an outermost periphery of the cover and defining a cover flange surface, the cover flange surface configured to biasingly engage a recessed surface of the recessed portion of the flange; and several magnets and several corresponding metal elements for disposition around the cover flange of the cover and around the recessed portion of the base ring to hold magnetically an entirety of the cover flange surface in biased engagement with an entirety of the recessed surface.

2. The apparatus of claim 1, further comprising: a cover insert extending forth from a cover side of the cover and forming an inner boundary of the cover flange surface, the cover insert being insertable into the base ring aperture, the cover insert defining an outer boundary having a frustoconical shape configured to guide the cover flange surface of the cover flange into biased engagement with the recessed surface of the flange by insertion of the cover insert into the base ring aperture.

3. The apparatus of claim 2, further comprising: a plurality of plates of planar triangular shape in planar radially aligned disposition around an outer surface of a barrel to form the cover insert with sides of the plates defining the outer boundary, vertexes of the plates at the outer boundary and the cover side define the inner boundary of the cover flange of the cover, the vertexes engage an edge of the recessed portion of the flange to align the cover flange surface in biased engagement with the recessed surface.

4. The apparatus of claim 1, further comprising: a sleeve detent formed at a sleeve end of the sleeve opposite the sleeve end of the sleeve attached to the flange.

5. The apparatus of claim 4, further comprising: an insert insertable into the sleeve, the insert configured to mechanically cooperate with the sleeve detent and to extend the base ring aperture to an ice fishing hole.
6. The apparatus of claim 1, further comprising: a handle in recessed disposition about a cover surface of the cover.
7. The apparatus of claim 1, further comprising: a tread formed in a cover surface of the cover.
8. An ice fishing hole cover apparatus, comprising: a base ring defining a base ring aperture and comprising a flange secured to a floor surface of a floor surrounding a floor hole formed in the floor by a plurality of fasteners and with a sleeve surrounding the base ring aperture at the flange extending from the flange into the floor hole; a recessed portion comprising an innermost portion of the flange, the recessed portion passing around an entire periphery of the base ring aperture and defining a recessed surface; a cover having a cover flange passing around an entirety of an outermost periphery of the cover and defining a cover flange surface, the cover flange surface configured to biasingly engage a recessed surface of the recessed portion of the flange; and several magnets and several corresponding metal elements disposed around the cover flange of the cover and around the recessed portion of the flange to hold magnetically an entirety of the cover flange surface in biased engagement with an entirety of the recessed surface.
9. The apparatus of claim 8, further comprising: a cover insert extending forth from a cover side of the cover and forming an inner boundary of the cover flange surface, the cover insert being insertable into the base ring aperture, the cover insert defining an outer boundary having a frustoconical shape configured to guide the cover flange surface of the cover flange into biased engagement with the recessed surface of the flange by insertion of the cover insert into the base ring aperture.
10. The apparatus of claim 8, further comprising: an insert inserted into the sleeve in mechanically cooperation with the sleeve detent to extend the base ring aperture to an ice fishing hole.